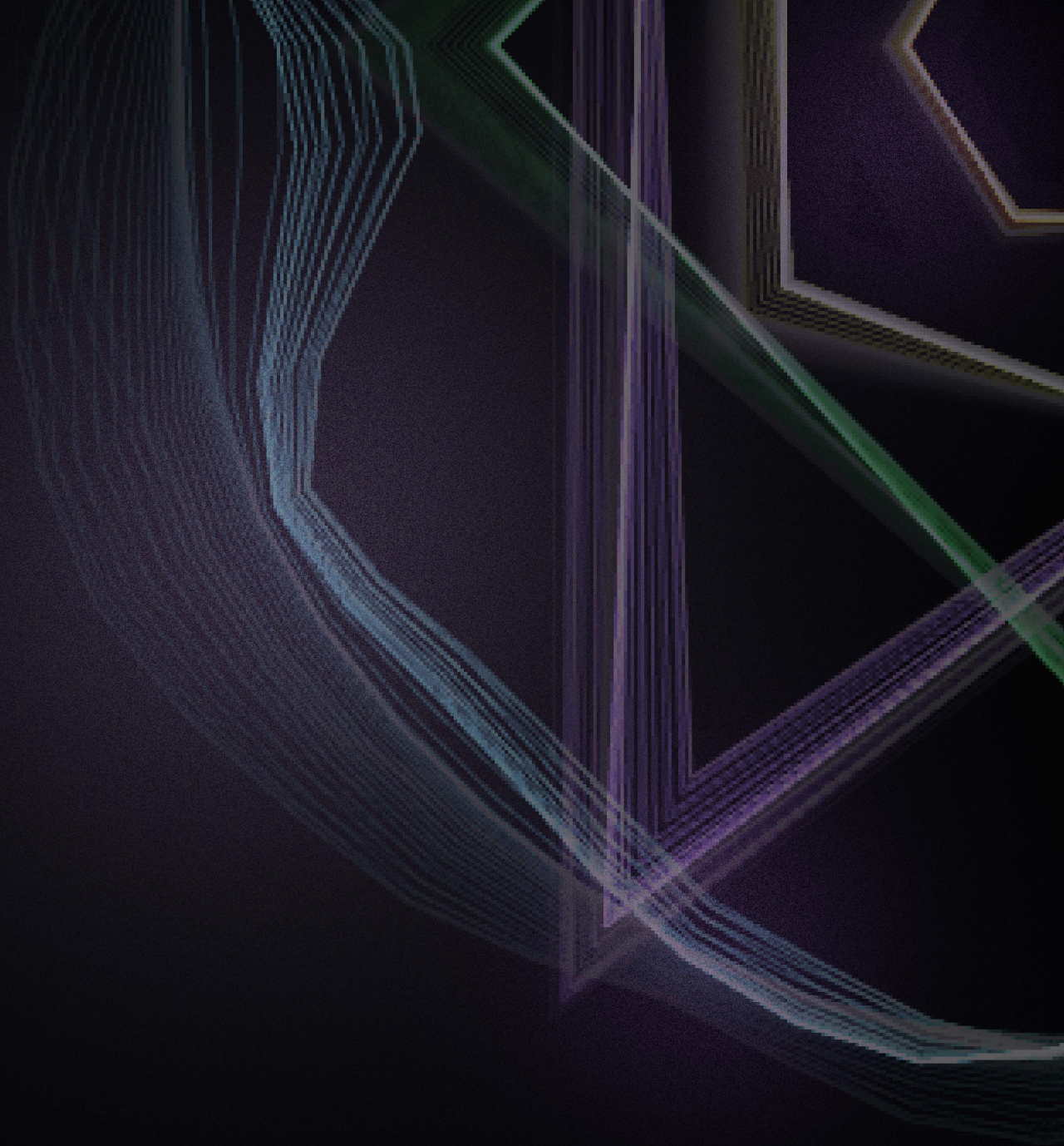


***SLEEP
WHISPER***



Leading in



Have you ever had this feeling after a nap?

▶ The more I sleep, the sleepier I get

Unable to wake up, or feeling groggy and extremely tired after waking up.



▶ I can't remember the dreams I had

After waking up, my mind is confused, and I just can't remember the dream I had.



Background research

77.1% OF PEOPLE TAKE NAPS REGULARLY.

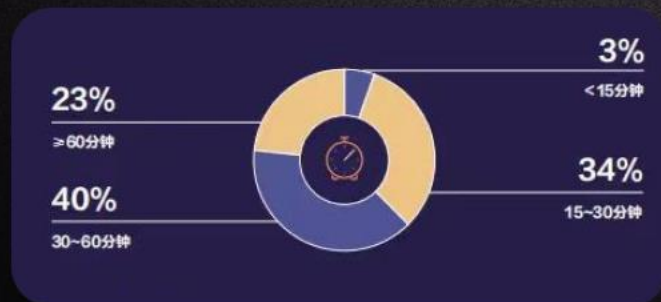
"Modern workers say, 'If you don't nap at noon, you'll collapse in the afternoon!' Spanish medical research shows that a 30-minute daily nap helps balance hormone secretion and reduces the incidence of cardiovascular diseases by 30%. According to a survey, 77.1% of people take naps, with 45.9% napping regularly, 31.2% napping occasionally, and 22.9% rarely taking naps."



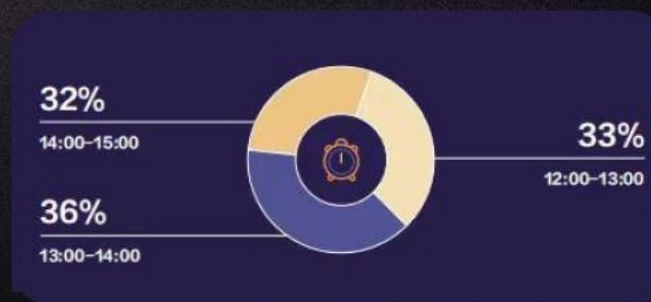
63% OF RESIDENTS NAP FOR MORE THAN 30 MINUTES.

"Residents generally have longer nap times, with 23% napping for over an hour. More than half of the people wake up from their nap after 2:00 PM. Sleeping too long during the day can reduce the quality of nighttime sleep."

NAP DURATION



NAP ONSET TIME



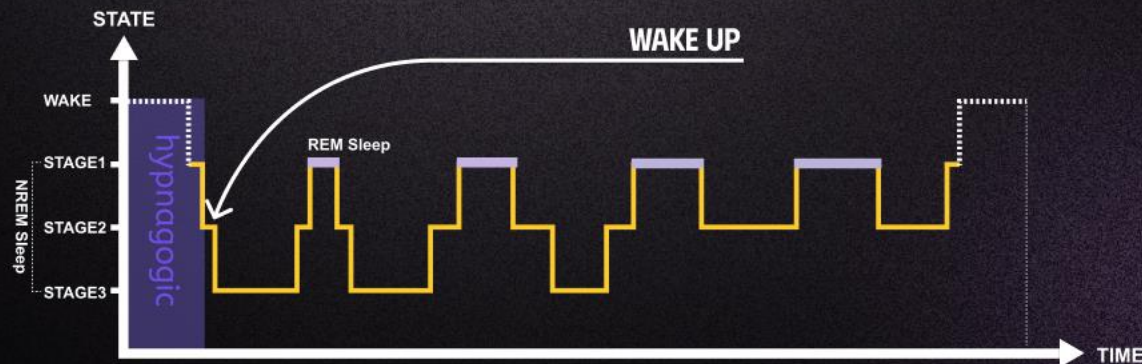
NAP WAKE-UP TIME



How to Take a Good Nap at Noon ?

"Life Cheat Code: Napping"

- ▶ **Memory Recovery:** A nap 'clears the cache' and restores hippocampal function, significantly improving memory."
- ▶ **Optimal Duration:** 10-20 minutes is best, avoid 30 minutes to prevent sleep inertia.

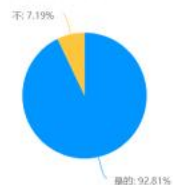


Questionnaire survey

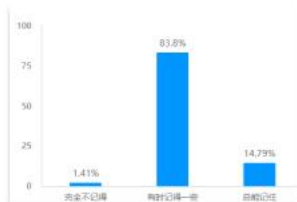
153

Number of completed questionnaires

1.你平时做梦吗？



2.你记得你的梦吗？



3.你能记得些什么？



4.你想捕捉记录你的梦境吗？



5.您有什么好奇的点或者意见或者建议吗？



▶ Most people **have limited memory of their dreams**, and the content of the dream may quickly fade after waking up.

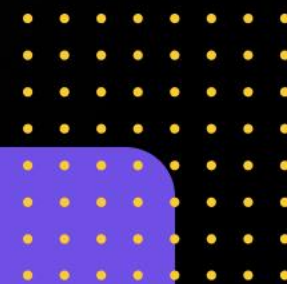
▶ Most people tend to remember the **overall emotions and scenes**, rather than the details.

▶ Most people are interested in interpreting, presenting, recording, and controlling their dreams, and they are generally curious about **how dreams are formed and their origins**.

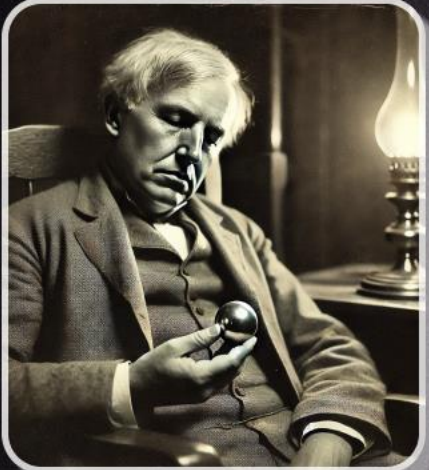
▶ **More than 70% of people** are interested in recording their dreams and would like to participate in trying it.



**Sleep well
Wake up well
Know about your Dream**



Hypnagogic State



Edison Steel Ball Experiment

When he just begins to fall asleep, his muscles relax, and the steel ball in his hand will drop to the floor, making a sound that wakes him up. At that moment, magical 'creative ideas' will pop into his mind, which can solve many complex problems.

MIT_Dormio: Interfacing with Dreams

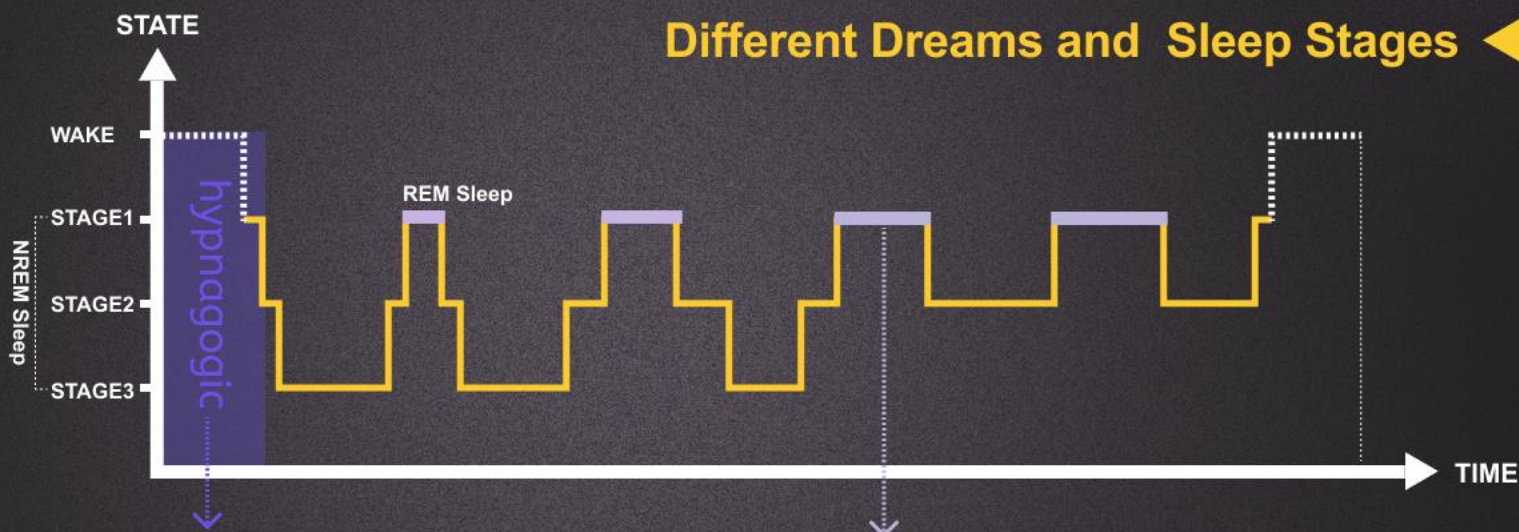


"The creator breaks free of logic and deductive reasoning, of familiar pathways, of taken-for-granted approaches."

- ▶ Micro-dreams can **inspire creativity**, promote unconscious thought connections, and help solve problems. Although dreams are vague, the subconscious connections made by the brain while processing information can offer new perspectives, helping to break free from fixed thought patterns and find unexpected answers.
- ▶ Micro-dreams usually occur during the early stages of sleep or in a semi-awake, semi-asleep state, when both the body and brain are relatively relaxed. Through brief micro-dreams, the brain can temporarily **detach from the stress and worries** of the day, helping to **regulate emotions and reduce anxiety**.
- ▶ During NREM sleep, the brain **still responds to external stimuli** like sounds, though these responses fade during REM sleep. When sounds occur in NREM sleep, they activate areas like the thalamus and auditory cortex, and this response can be stronger when they trigger K-complexes. People frequently experience hypnagogic states may have **heightened brain responses to these stimuli**.

Sleep Stages Detect & Classify

Different Dreams and Sleep Stages



MICRODREAMS

They are usually very brief, lasting only a few seconds, and have a simple structure, often consisting of just one or a few sensory elements. The emotions involved are minimal, and the memories may be more closely related to recent experiences from wakefulness. These dreams are often isolated, discontinuous experiences, and are more vague, lacking clear visual details.

SPARK!

COMMON DREAMS

They last longer and contain more complex and detailed plots. Typically more intricate, they include rich storylines, characters, and emotions. They may involve a broader range of memories, including distant past experiences and abstract concepts. After waking up, they are often easier to recall and remember.

STORY!

How to Detect Sleep

Polysomnography (PSG)



ECG

EOG

EMG

Breathe

SpO2

EEG

ECG

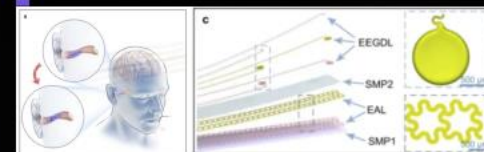
HRV

ST

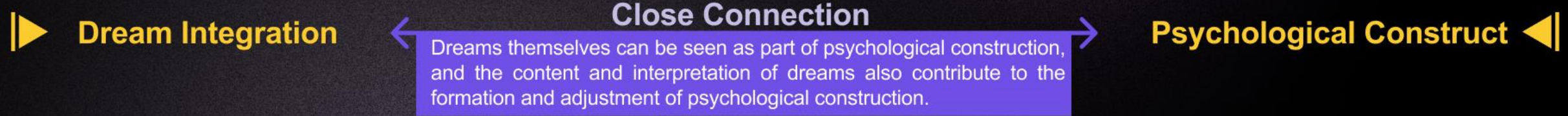
SpO2

Motion

Wearable Device



Dream Integration & Psychological Construct



Dream Integration is a complex concept involving psychology and cognitive science. It generally refers to the process during sleep, especially in the rapid eye movement (REM) sleep stage, where the **brain reorganizes and integrates daytime experiences, memories, emotions, and cognitive processes.**

Psychological construction refers to the way individuals **use thinking, emotions, and behaviors to understand themselves, others, and the world.** It focuses on how individuals build their understanding of the world, interpret their emotions, and respond to life challenges based on these understandings.

THE ROLE OF DREAM INTEGRATION

1

Memory Consolidation: During sleep, the brain transfers information from short-term to long-term memory, improving learning and memory.

2

Emotional Regulation: Dreams help process and release negative emotions, reducing anxiety, stress, or trauma.

3

Cognitive Integration and Problem Solving: Dreams reorganize daily problems, aiding unconscious creative processing.

TEXT
DESCRIBE

BIO-DATE
VISUALISATION

ABSTRACT
ART

HOW TO RECORD DREAM

Expressive therapies use creative forms of expression, like **art** and **writing**, to **help individuals process emotional experiences** and improve mental health. These therapies have been effective for conditions such as nightmares, trauma, and depression. The goal is to help clients discover personal meaning, leading to greater self-understanding, emotional healing, conflict resolution, and overall well-being.

Metaphor: Ambiguous Narratives

How to Record Dreams Based on EEG Data

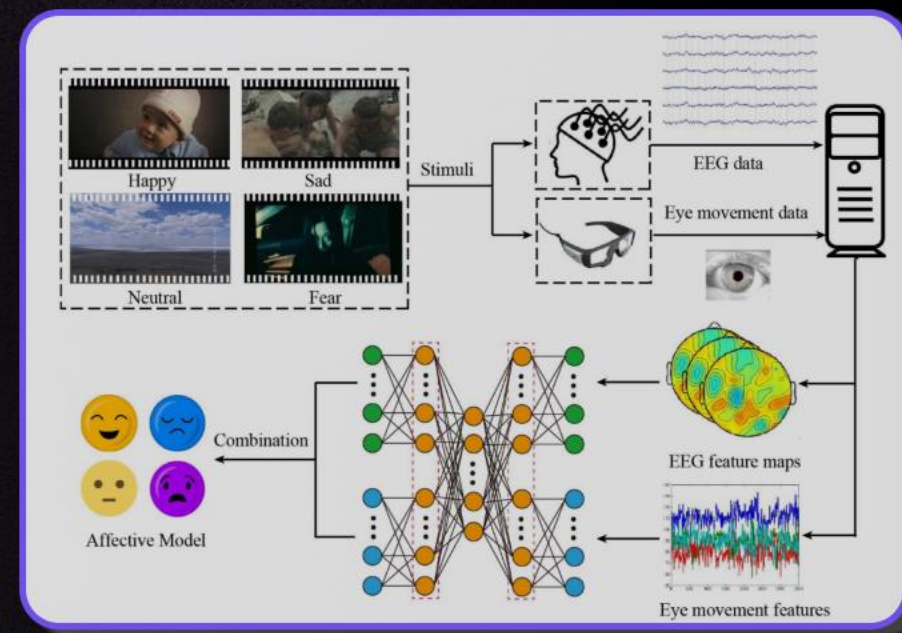
► Transform EEG to Emotional Dimension

Popular public affective computing datasets

Dataset	Description	Task Model
MAHNOB-HCI	27 subjects' EEG and various physiological signals, images, and video information	VAD (Valence-Arousal-Dominance) model
RECOLA	46 subjects' ECG, EOG, and audio, video information	VA (Valence-Arousal) model
DECAF	30 subjects' EOG, ECG, EMG, and video information	VA model
ASCERTAIN	58 subjects' EEG, ECG, GSR, and image information	5 emotion categories, VA model
AMIGOS	40 subjects' EEG, ECG, GSR, and image, video information	5 emotion categories, VA model
DREAMER	23 subjects' EEG, ECG	VAD model
RCLS	14 subjects' EEG	3 emotion categories
MPED	23 subjects' EEG, ECG, RSP, GSR	7 emotion categories
HR-EEG4EMO	27 subjects' EEG	Happy, Sad emotion categories
SEED	15 subjects, each with 3 sessions of EEG	3 emotion categories
SEED-IV	15 subjects, each with 3 sessions of EEG, EM	4 emotion categories
DEAP	32 subjects' EEG, EOG, EMG, GSR, RSP, BP, ST	VAD model

Human emotional changes are often reflected in physiological signals, which can more accurately represent emotions than facial expressions or speech, which are easily disguised. Common physiological signals for emotion computation include EEG, eye movement, EMG, skin conductance, ECG, and respiration. These signals typically have low frequencies and can be affected by the external environment, requiring specialized equipment and preprocessing to improve signal quality and computation effectiveness.

SEED-IV DATABASE



The SEED-IV dataset contains EEG and **eye movement data** of 12 subjects and **EEG data** of another 3 subjects. Data was collected when watching film clips. The number of categories of emotions changes to: **happy, sad, fear, and neutral**.

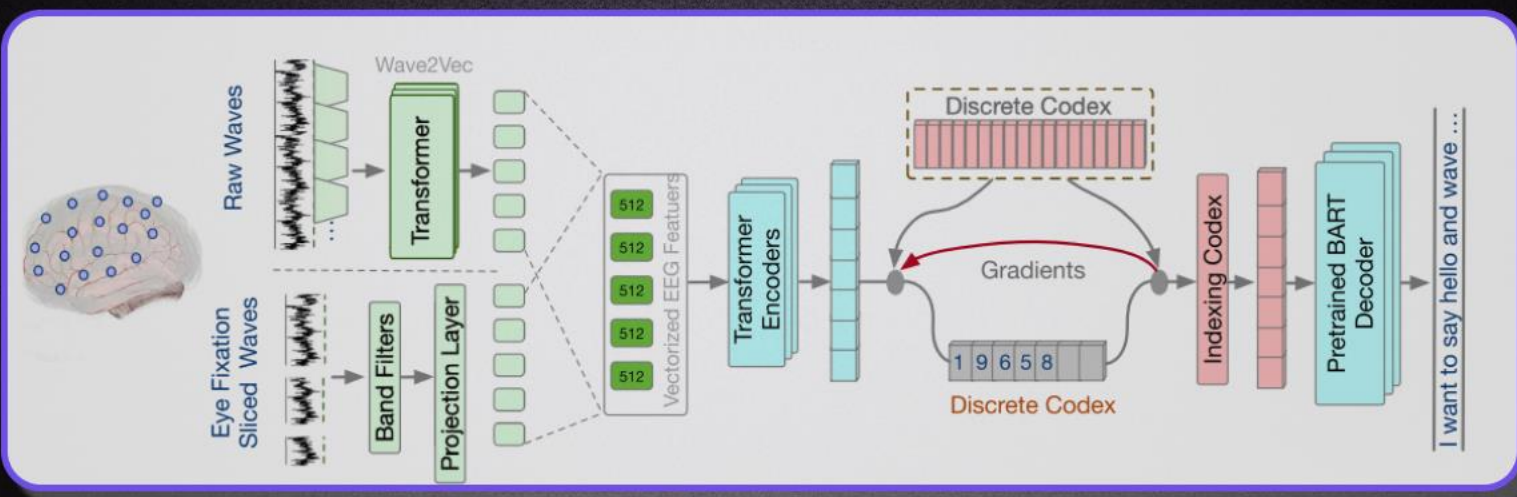


How to Record Dreams Based on EEG Data

CHINESEEEG DATABASE

► Transform EEG to Semantic Content

DEWAVE: can be generalized for both word-level EEG features and raw EEG wave translation.

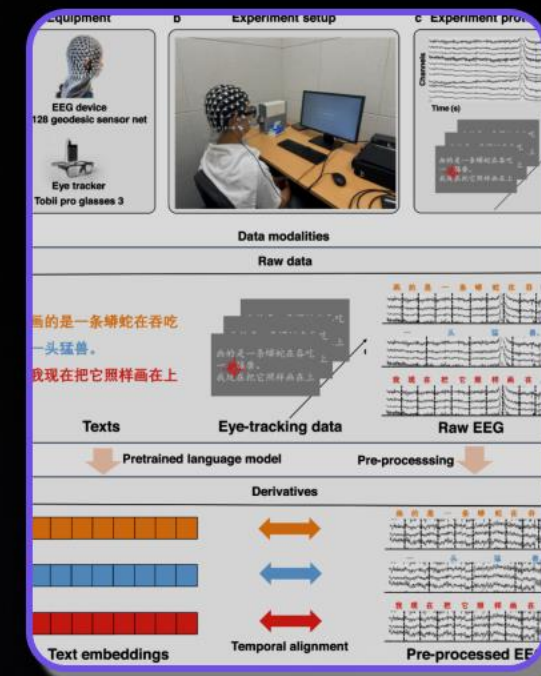


This paper introduces discrete codex encoding to EEG waves and proposes a new framework, DeWave, for **open vocabulary EEG-to-Text translation**.

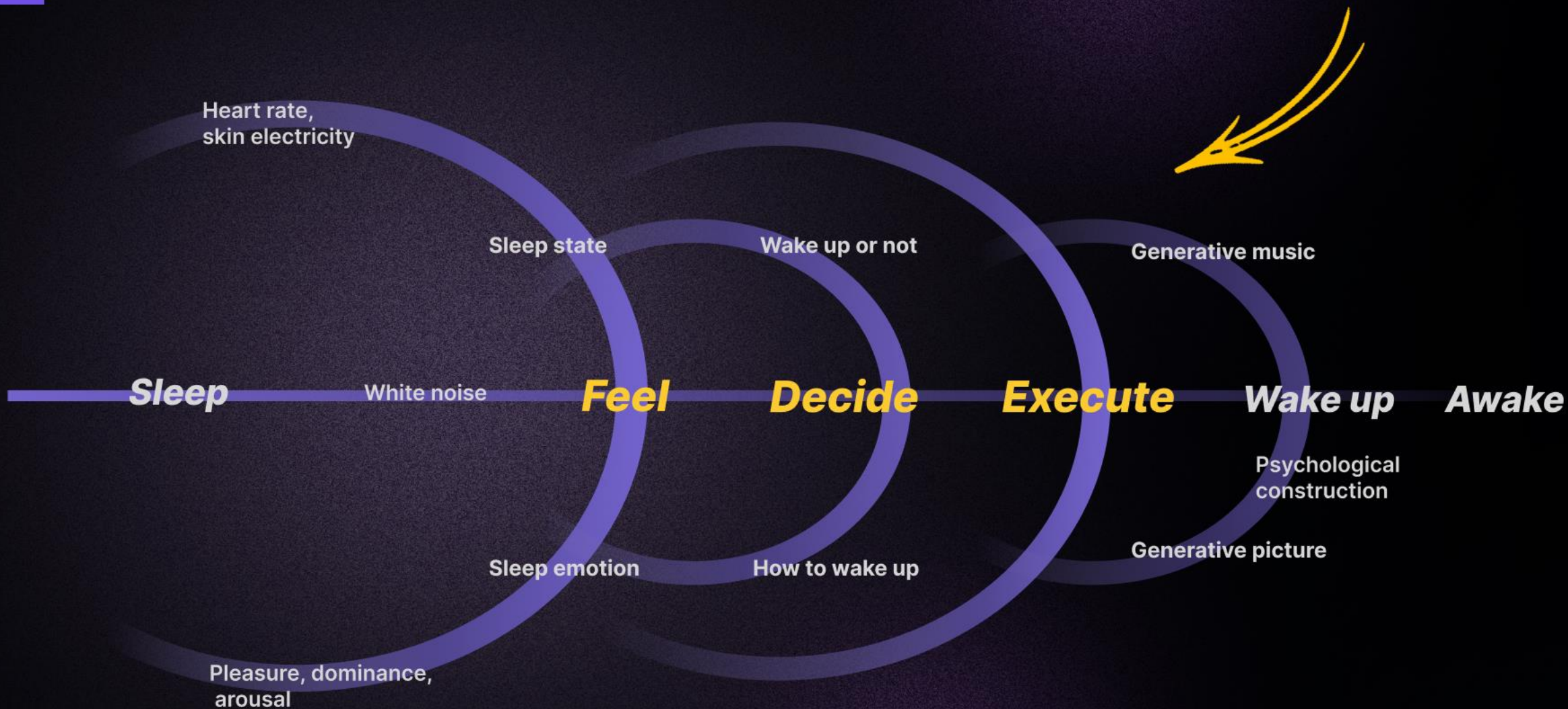
By utilizing **discrete codex**, DeWave is the first work to realize **the raw EEG wave-to-text translation**, where a self-supervised wave encoding model and contrastive learning-based EEG-to-text alignment are introduced to improve the coding ability.

Experimental results suggest the DeWave reaches SOTA performance on EEG translation, where it **achieves 41.35 BLUE-1 and 33.71 Rouge-1**, which outperforms the previous baselines by 3.06% and 6.34% respectively.

The “ChineseEEG dataset” is a high-density EEG dataset that integrates eye-tracking recordings, aimed at addressing the scarcity of EEG datasets with Chinese linguistic stimuli. As an EEG dataset derived from natural Chinese language stimuli, “ChineseEEG” supports research in neuroscience, natural language processing, and linguistics, establishes a benchmark for Chinese semantic decoding, advances the development of brain-computer interfaces (BCI), and contributes to exploring the **alignment between language models and human cognitive processes**.



Functional Framework



Form creativity



Headphones

+

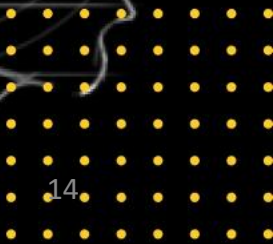


Neck pillow

= ?



Sk



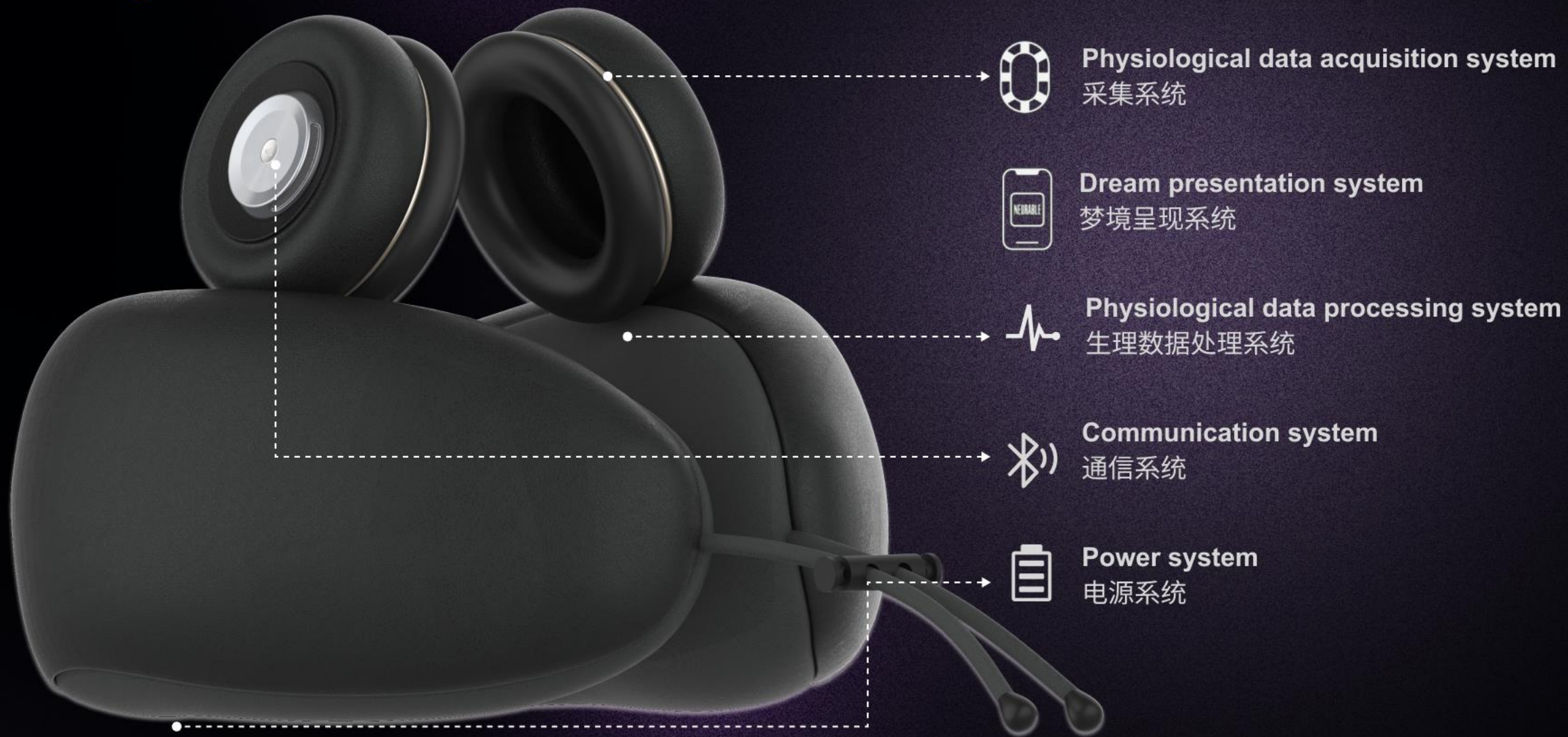
一枕， 解决你的午睡问题

A pillow solve all your napping problem

搭载EEJ脑电设备的sleep whisper支持脑电记录功能，随着你梦境内容的不断变化，它可以自动选取更需要你留意的数据，为你记录梦境数据。通过顺畅地融合主动降噪和通透模式，自适应音频可在各种环境下，为你调节降噪程度，带给你全新的午睡体验。



Hardware Design

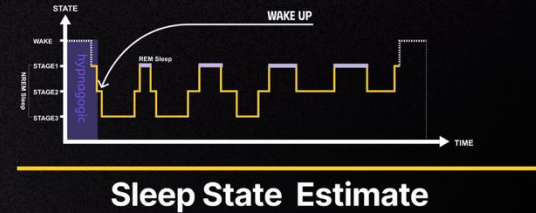


Physiological data acquisition & processing system



Electrocardiogram (ECG)
Heart Rate Variability (HRV)
Inertial Measurement Unit (IMU)
Peripheral Capillary Oxygen Saturation
Electroencephalogram (EEG)

Data Acquisition

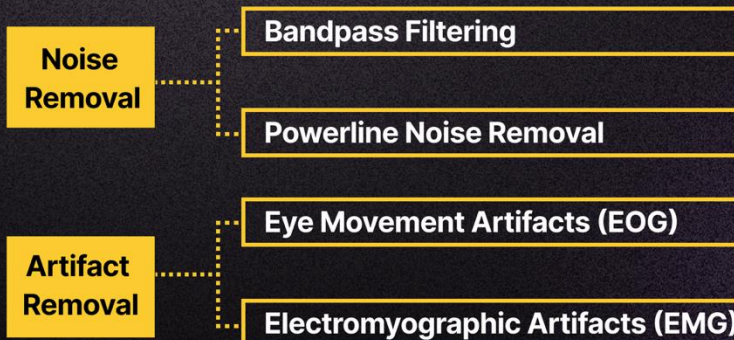


WAKE UP ?

STEP 1 Data Preprocessing

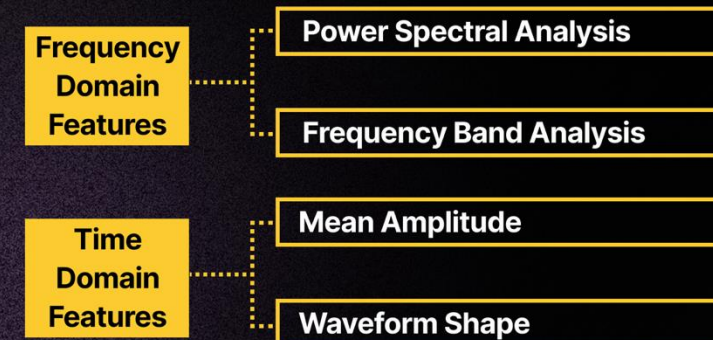
The goal of preprocessing is to **remove noise and artifacts**, making the signal clearer and more reliable

Date processing



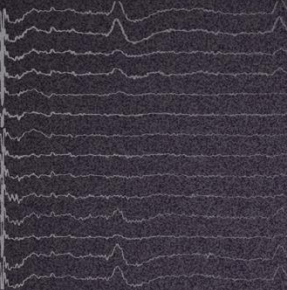
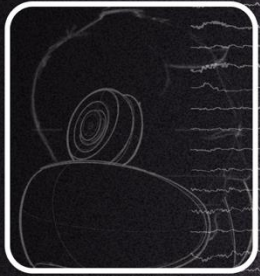
STEP 2 Feature Extraction

Feature extraction involves **extracting useful indicators** from the preprocessed EEG signals that are relevant to the research questions.



Dream presentation system

Graph Generation



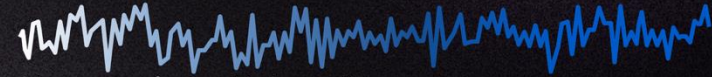
δ 0.1-3 Hz

θ 4-7 Hz

α 8-12 Hz

β 12.5-28 Hz

γ 29-50 Hz



Unconscious, Relaxed



Meditation, Creative thinking, Emotional processing



Calmness



Concentrate, Exciting



Advanced cognition, Information integration

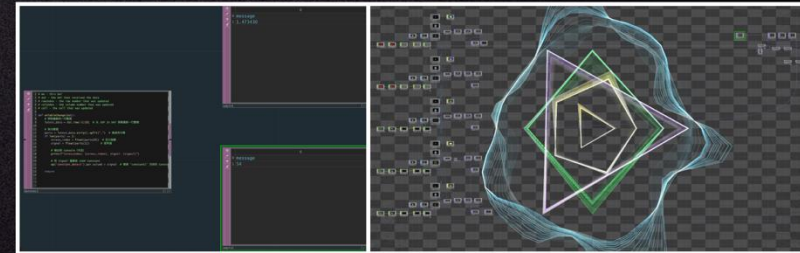


Power Spectral Analysis

Fast Fourier Transform (FFT) or Wavelet Transform is used to convert EEG signals into the **frequency domain**, analyzing the power in different frequency bands (such as delta, theta, alpha, beta waves).



Blue tooth



Raw Data

Remove Noise
Power Spectral Analysis

EEG signals into the different frequency domain back
to the UDP element in TD to achieve visualisation.

Dream presentation system

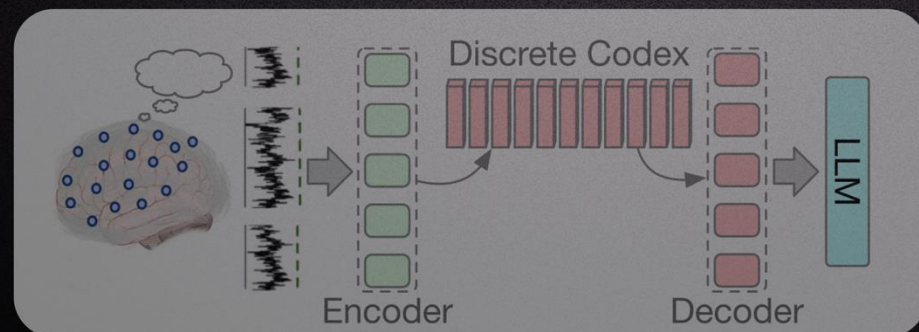
Text Generation



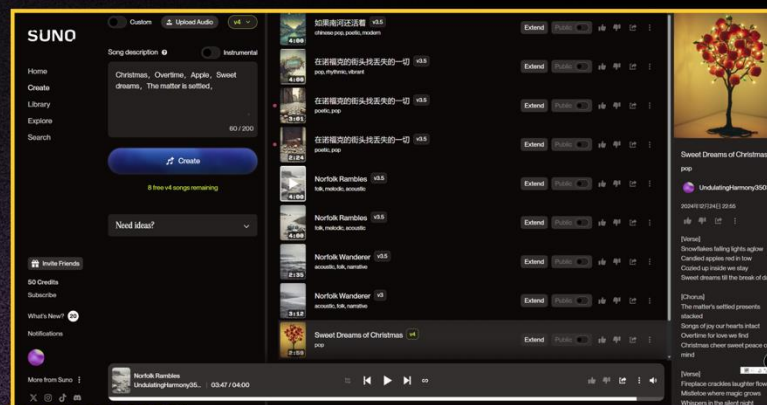
Transform EEG to Emotional Dimension



Transform EEG to Semantic Content



Enrich visual elements

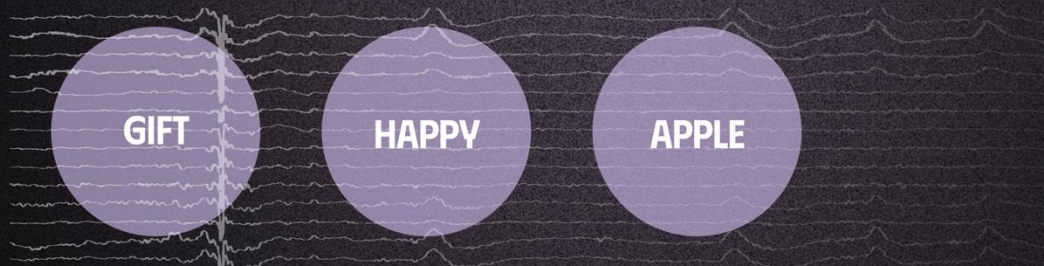


Generate alarm songs based on keywords.

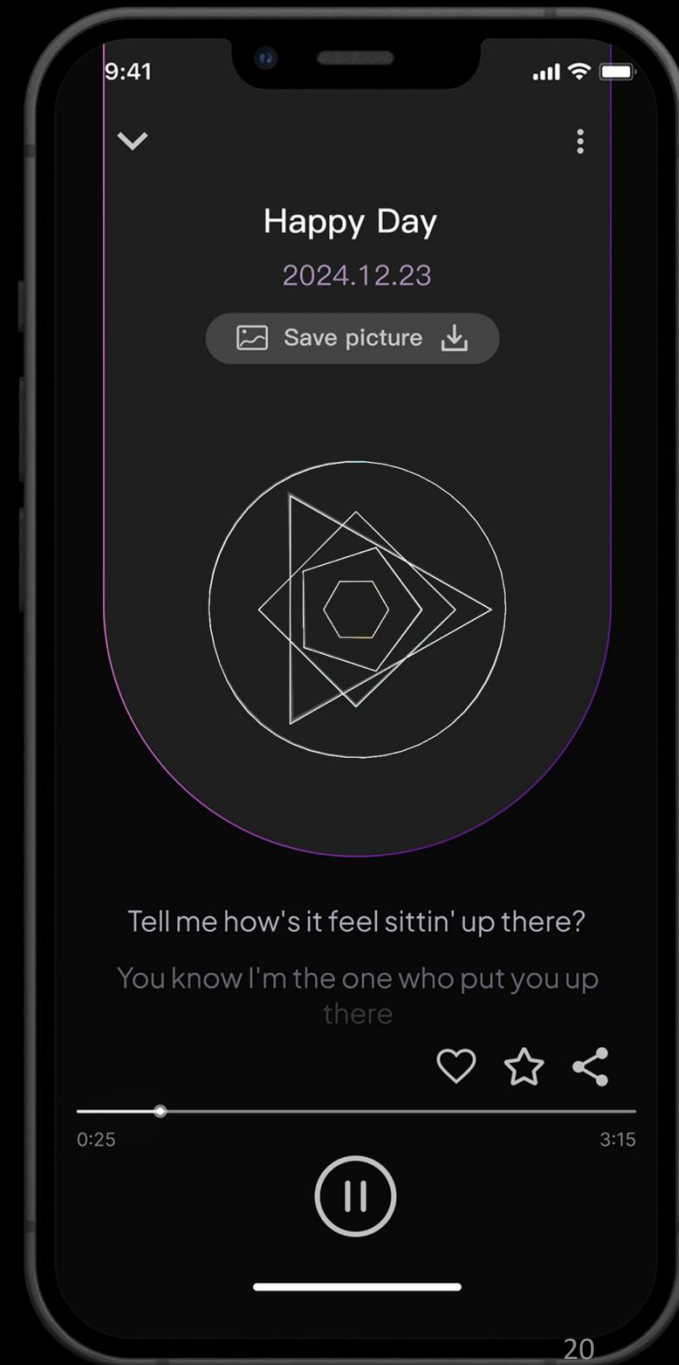
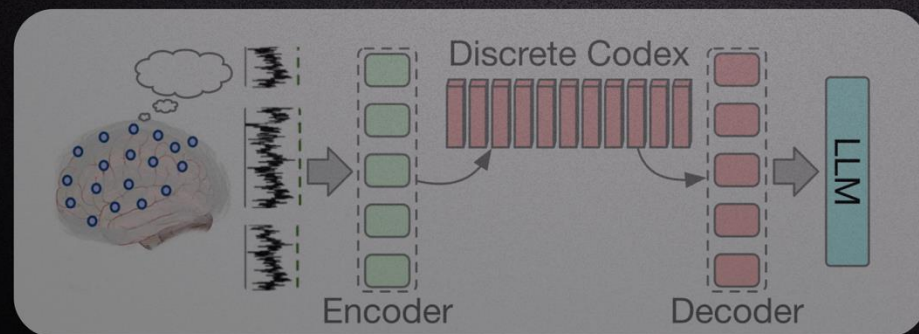
Dream presentation system



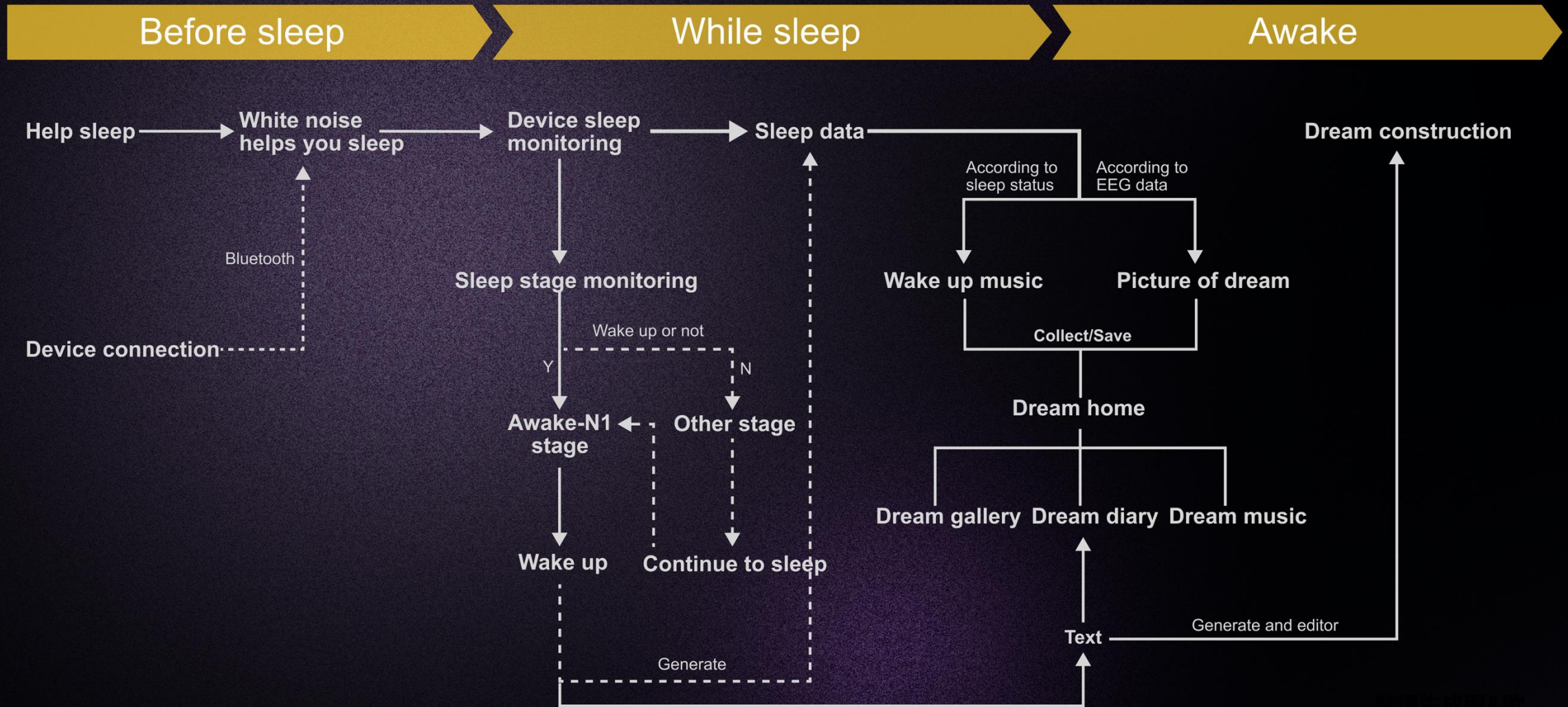
Transform EEG to Semantic Content



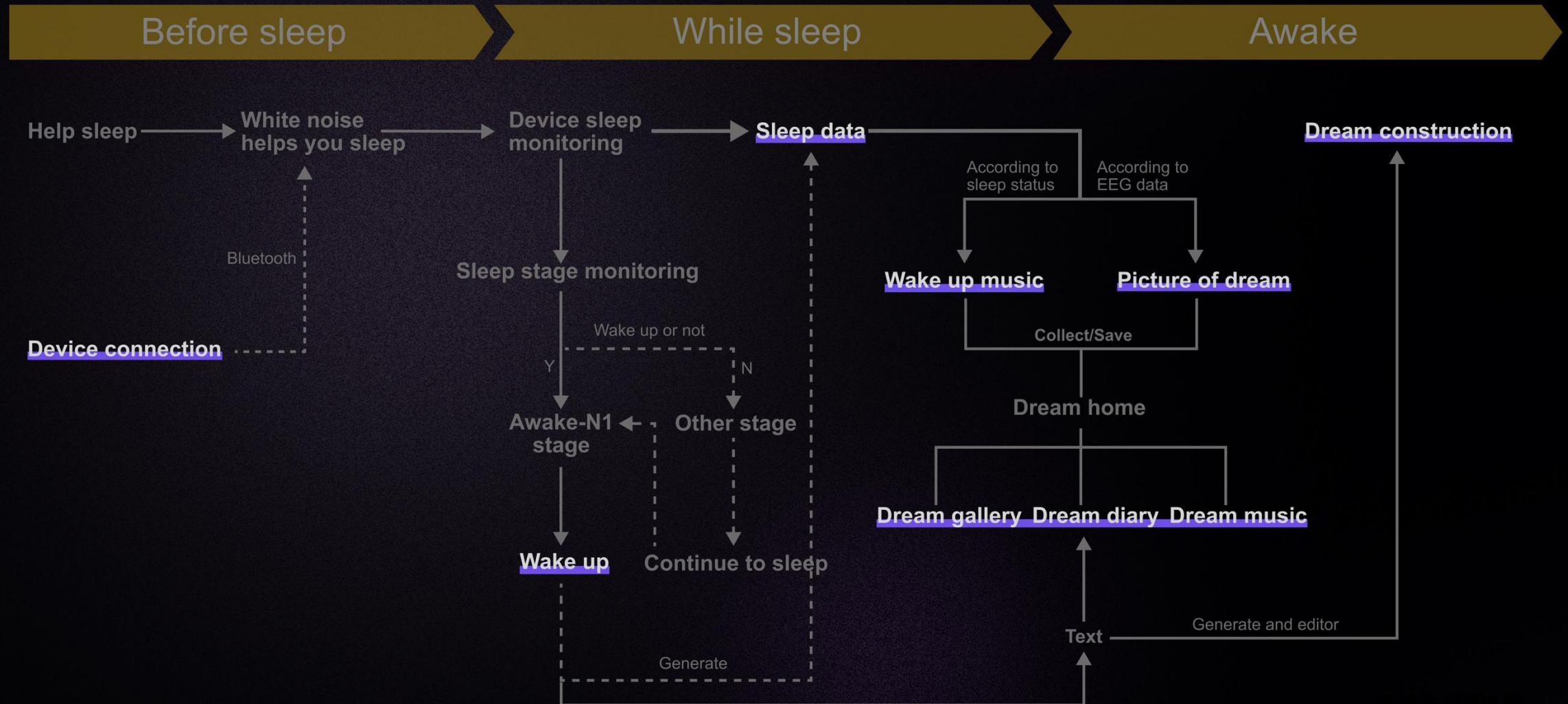
Transform EEG to Emotional Dimension



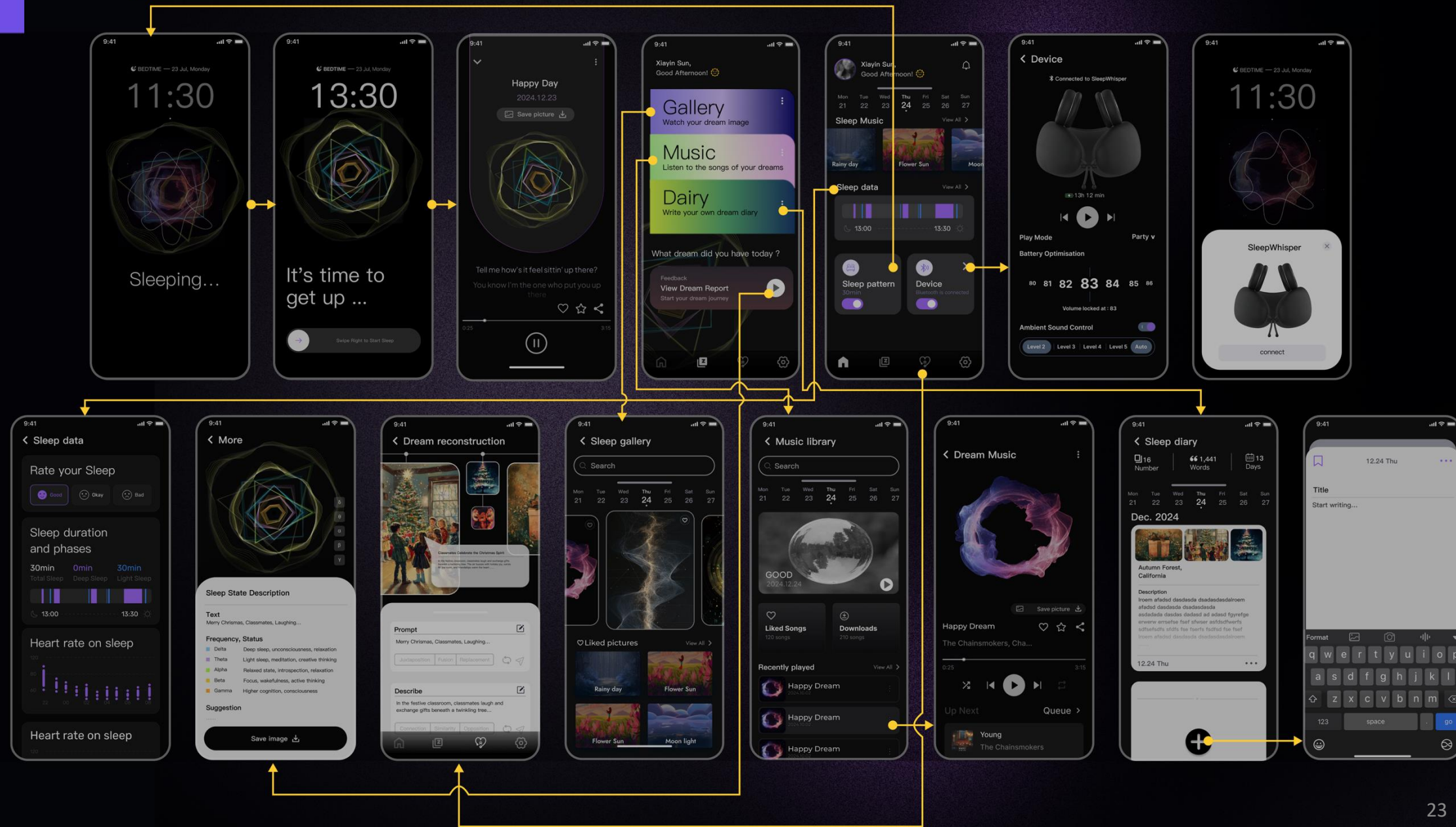
Core processes



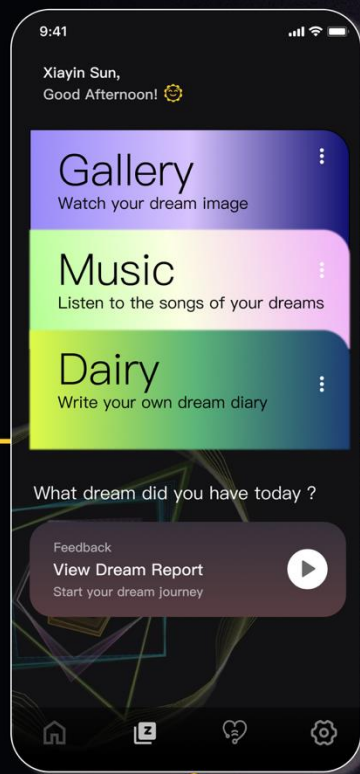
Core processes



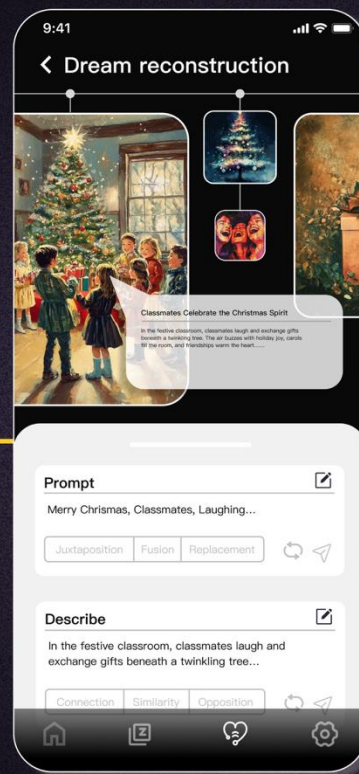
UI Design



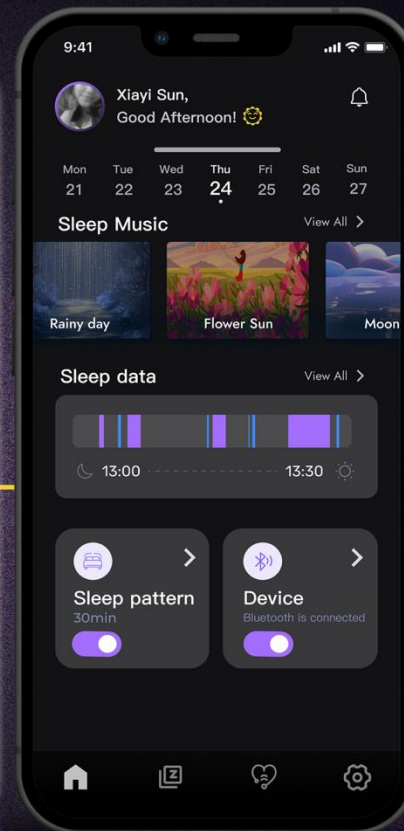
Interface introduction



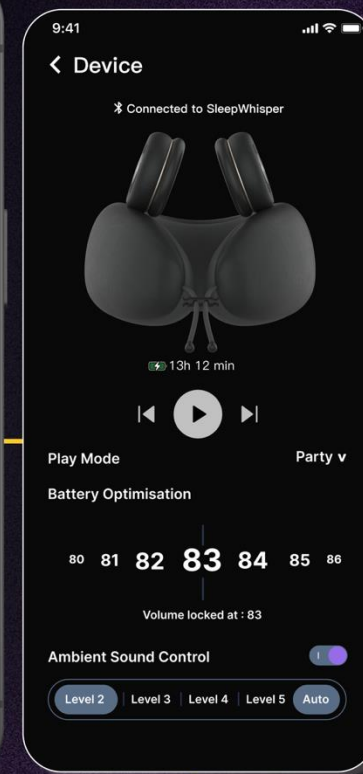
Dream home



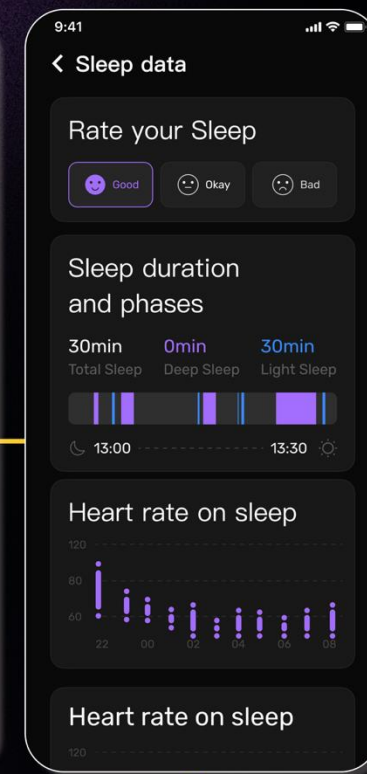
Dream reconstruction



Main interface



Equipment control



Data monitoring

Use process

Before sleep



Use process

While sleep

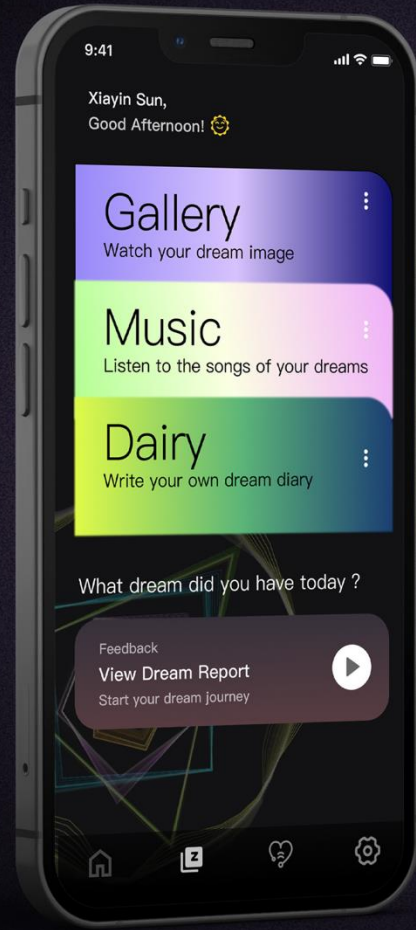
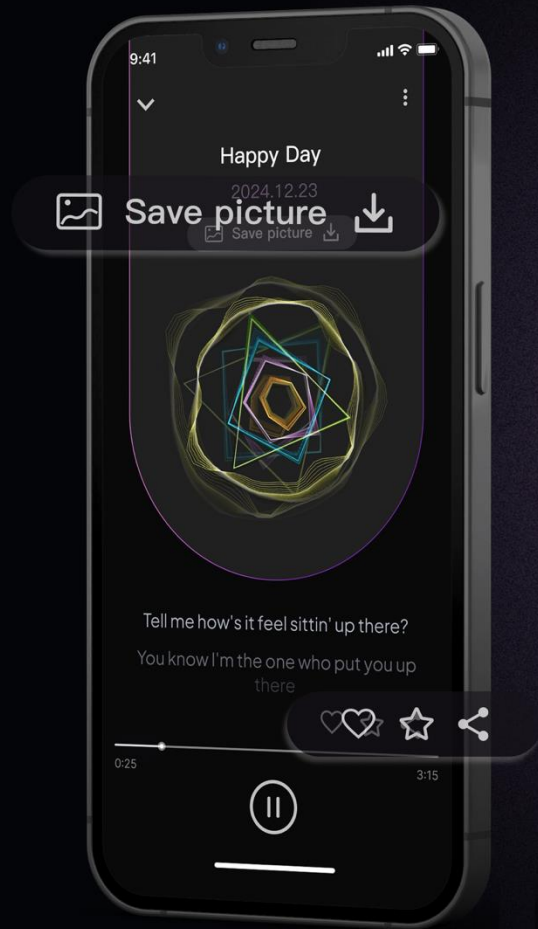


Wake up according to sleep stage

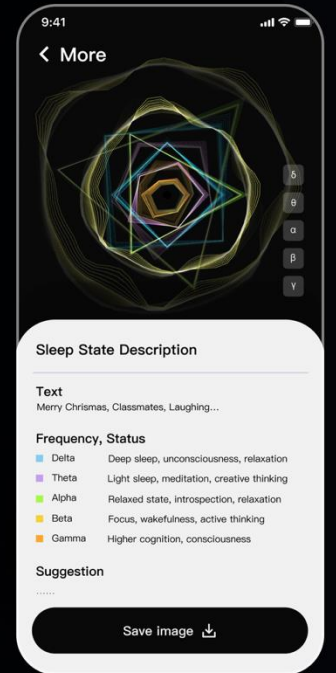
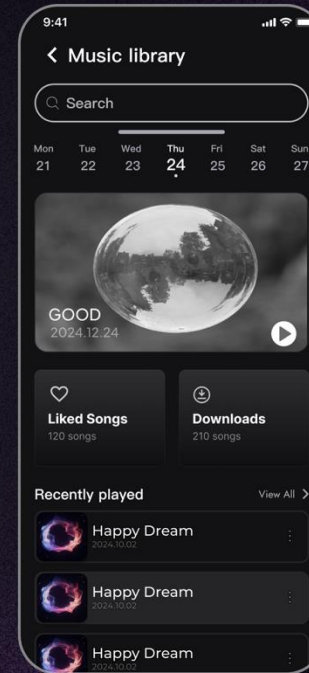
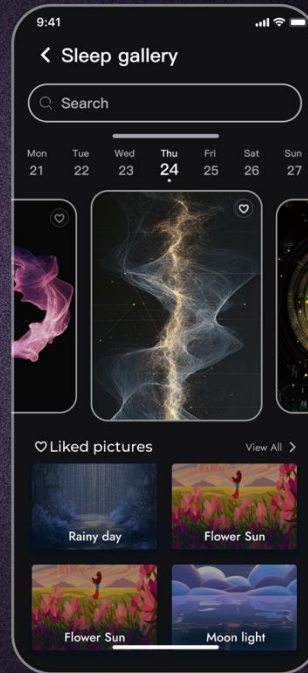
Generate wake up music and images
based on sleep states



Use process Awake



Your Dream Home



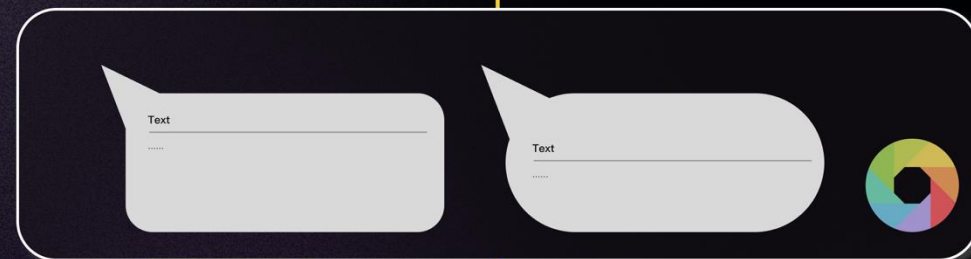
Gallery

Music

Dairy

Report

Use process Awake

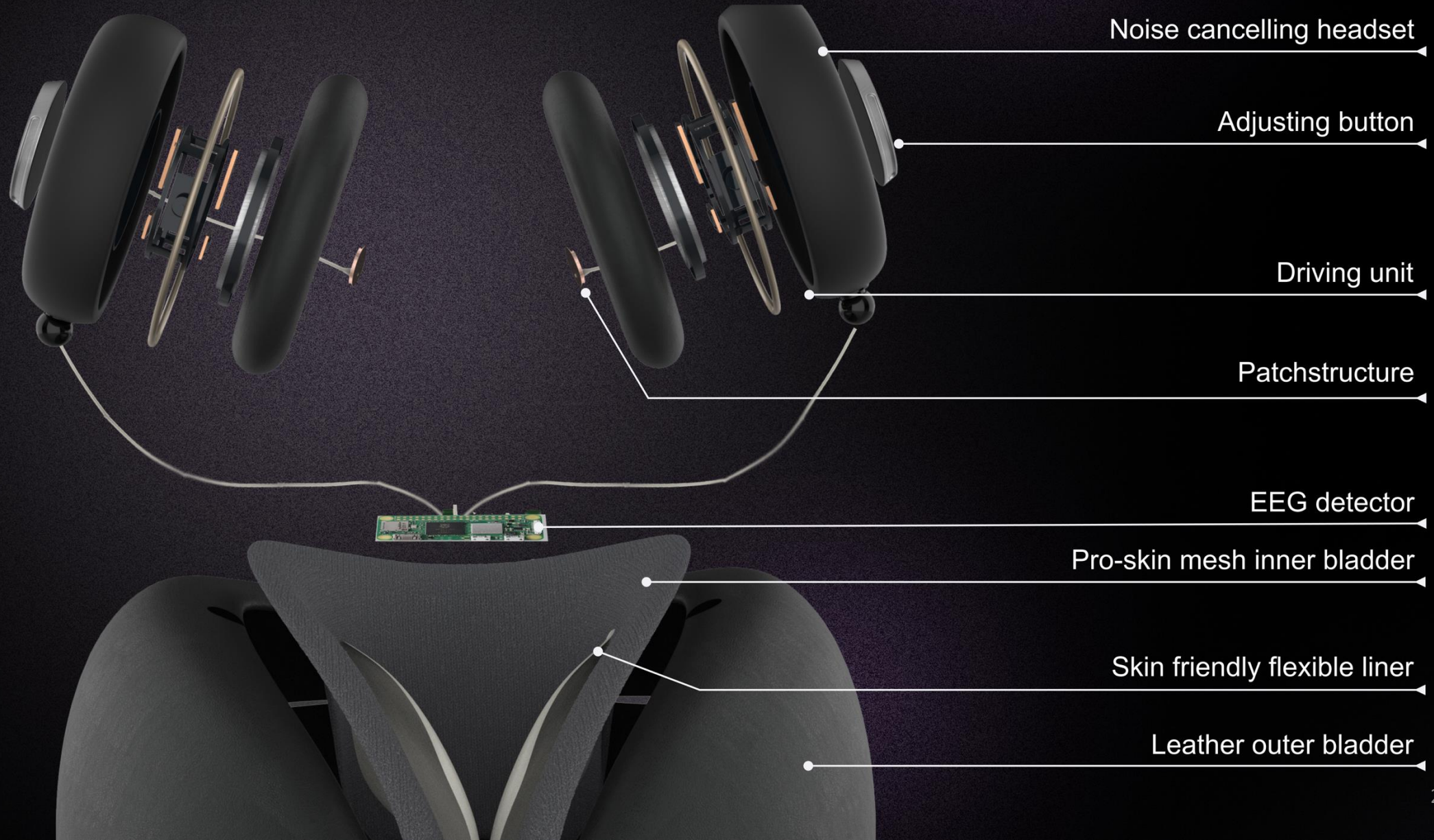


The corresponding image
is generated from the generated Text

Autonomous definition

The user recalls the dream to
arrange the storyline through emotional arcs

Explosion diagram



Use process
Equipment control



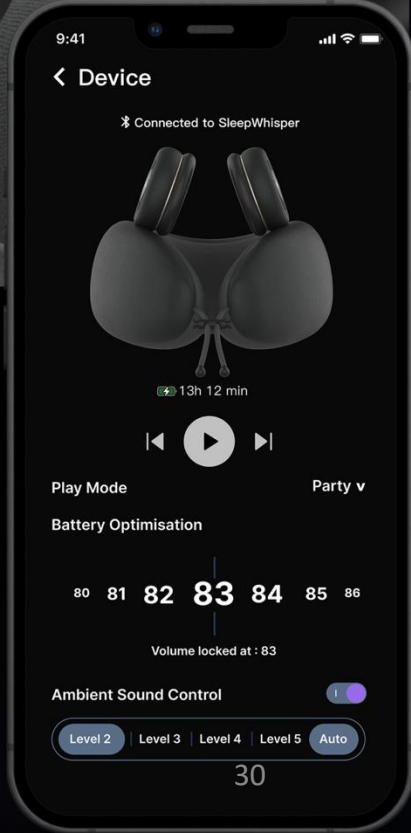
Bluetooth connection

Rotate

Volume adjustment

Music play/pause

Type - C charging port



The advertisement features a dark background with two large, stylized, rounded shapes representing the product. The left shape is dark grey/black, and the right shape is white. Both shapes have a gold-colored band across the middle. Above the text, there are two small, horizontal, rounded rectangular shapes, one dark grey and one white, separated by a thin vertical line. The text "Sleep Whisper" is centered in a large, white, sans-serif font. Below it, the text "A variety of colors to choose from" is centered in a smaller, white, sans-serif font.

Sleep Whisper

A variety of colors to
choose from

Catch Your Dream

Sleep well / Wake up well
Know about your Dream

进一步了解

购买



THANKS!